$$F(x, y, z) = 2x^2 - y^2 - z^2$$

Let $\vec{r} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$$F(\vec{r}) = \vec{r}^{\,T} M \vec{r}$$

where $M = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

$$\nabla F = \frac{\partial}{\partial \vec{x}} (\vec{x}^{\,T} M \vec{x}) = 2M\vec{x}$$

Substituting $\vec{x} = \begin{bmatrix} 1 \\ 0 \\ \sqrt{2} \end{bmatrix}$ for point P,

$$\nabla F = 2 \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ \sqrt{2} \end{bmatrix}$$

$$\nabla F = \begin{bmatrix} 4 \\ 0 \\ -2\sqrt{2} \end{bmatrix}$$

The gradient vector ∇F at a point is always $\perp$ to the surface.

$$\|\nabla F\| = \sqrt{16 + 8} = 2\sqrt{6}$$

$$\therefore \vec{n} = \frac{\pm \nabla F}{\|\nabla F\|} = \begin{bmatrix} \pm\sqrt{2/3} \\ 0 \\ \mp\frac{1}{\sqrt{3}} \end{bmatrix}$$